INCH-POUND

MIL-STD-657B
28 August 2025
SUPERSEDING
MIL-STD-657A
28 July 1978

DEPARTMENT OF DEFENSE
STANDARD PRACTICE

PROVISIONS FOR EVALUATING QUALITY

OF

SERVICE CAPS

AMSC N/A

FSC 8405

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FOREWORD

1. This standard is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

2. This standard is intended to provide standard practice for quality inspections of service caps. This standard provides the following information:

- A standard method for visually inspecting and evaluating the quality of service caps.
- Standard descriptions for service caps defects.
- Standardized examination requirements for service caps.

3. Comments, suggestions, or questions on this document should be sent to trpsptctstandardiz@dla.mil. Since contact information can change, you may want to verify the currency of this address information using ASSIST online database at https://assist.dla.mil.

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1. SCOPE

1.1 Scope. This standard covers the inspection provision requirements for service caps worn by members of the military.

2. APPLICABLE PUBLICATIONS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE HANDBOOK

MIL-HDBK-156 – Military Standardization Handbook – Glossary of Military Clothing Fabrication Terms

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-3095 - Caps, Service, Man's, Air Force

(Copies of this document are available online at https://quicksearch.dla.mil.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Photographic Seam Smoothness Scales - 68373A

Scales are available online at https://aatcc.org).

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org).

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ASTM INTERNATIONAL

ASTM D3990 - Standard Terminology Relating to Fabric Defects
ASTM D6193 - Standard Practice for Stitches and Seams

(Copies of these documents are available online at https://www.astm.org).

OTHER PUBLICATIONS

Fabric Replica Defect Scales
Part Number 402985 – Slub/Knot Replica Set

(Replica set is available for purchase from SDL Atlas Customer Service, 1-803-329-2110).

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document or end-item specific documents supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 End-item. End-item refers to finished service caps ready for inspection and acceptance.

3.2 Fabric defects. Fabric defects specified in Table I are defined in ASTM D3990. Fabric defects not specified in Table I are specified in the appropriate fabric specification.

3.3 Inside and outside. The terms “inside” and “outside” as used in this standard are defined as follows:

Inside - Any part of the service caps that is not visible when the caps is examined from 3-feet at shoulder level height, positioned as worn, and be completely rotated.

Outside - Any part of the service caps that is visible when the caps is examined from 3-feet at shoulder level height, positioned as worn, and be completely rotated.

3.4 Knots and slubs. Score only knots and slubs that exceed the limits specified in the basic cloth specification as defects when evaluating quality of the finished service caps. Knots and slubs are rated using Fabric Defect Replica Scales (see 2.3).

3.5 Major defect. A departure of a quality characteristic from its intended level of the specification that occurs with sufficient severity that is likely to result in a failure, or to reduce considerably the usability of the service caps for its intended purpose. (See Table I).

3.6 Minor defect. A departure of a quality characteristic from its intended level of the specification but the service caps is still considered serviceable. (Multiple minor defects may render service caps or production lot unserviceable). (See Table I).

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TABLE I. End item defects.

Defect Description	Defect Category	
	Major	Minor
I. MATERIAL DEFECT AND DAMAGE:		
Any hole, cut, tear, smash, burn, drill hole, scorched area and/or multiple floats on outside and inside (see 3.3) (longest dimension in any direction):		
- up to 1/4 to 1/8 inch, inclusive		201
- more than 1/4 to 1/8 inch	101	
Any knot and/or slub (see 3.4) 1/		202
Misweave, area of poor/no dye (print) penetration, dye streak, missing yarn, broken yarn, visible mend, thin place, shade bar and barre		203
Loose yarn, snag and/or coarse yarn more than 1/4 to 1/8 inch in any direction on outside	102	204
Woven in debris and/or waste on outside visible at 3-feet		
Permanent fold, pleat, or crease in cloth (when caught in seaming or stitching) on the outside	103	
Faux Leather:		
- cracked, cut or thin, weak spot	104	
- spongy, flanky or stretched (showing a coarse grain and uneven appearance)		205
- any scar, wrinkle or scratch visible at 3-feet		206
- finish on faux leather peeling off	105	
- needle chews in faux leather		207
II. SHADED PARTS		
Variation in shade within a part or between parts	106	
III. STITCHING AND SEAM DEFECTS		
Accuracy of seaming:		
- seam puckered (score only when on major portion of seam) (see 3. 7)		208
- seams irregular, twisted, wavy or pleated	107	
- part caught in any unrelated operation or stitching		209

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TABLE I. End item defects. – Continued

Defect Description	Defect category	
	Major	Minor
III. STITCHING AND SEAM DEFECTS - (cont'd)		
Accuracy of seaming:		
- end of stitching when not caught in another seam or stitching backtacked less than 1/4 to 1/8 inch; or end of continuous stitching overlapped less than 1/2 inch		210
- stitching overlapping end of thread break less than 1/2 inch		211
- not specified color or varying shades of thread used on outside	108	
Gage of stitching and seam allowance:		
- edge, top or raised stitching sewn too close to edge resulting in damage to cloth		212
- seam allowance not as specified by 1/16 inch or more (score only when condition exists on more than half the length of the seam)		213
- irregular or not within range specified or varies more than 1/16-inch when no range is specified. Score only when condition exists on major portion (i.e., more than half the length) of the seam		214
Raw edge: (see 3.9)	109	
On outside		
- up to 3/4-inch inclusive	110	
On inside: (excluding left end of curtain):		
- 1/4 to 1/8 inch up to and including 1/2 inch	111	
- more than 1/2 inch up to and including 1 inch	112	
- more than 1 inch (not including inside lining bottom edge)		215
- raw edge on inside lining along bottom edge more than 1 inch		2/
Seam and stitch type:		
- wrong seam and/or stitch type	113	
- any line of stitching omitted	114	
- any line of stitching not starting and/or ending were specified		217
Basting stitch visible on outside of the finished caps	115	
Looper thread of 401 stitch type on outside		218
Stitches per inch (score only when condition exists on more than half the length of seam):		219
- less than the minimum by two (2) or more stitches		220
Stitch tension:		
- loose stitch tension resulting in a loose seam		221
- loosely exposed loops of lower or top thread on edge, top or raised stitching		222

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TABLE I. End item defects. – Continued

Defect Description	Defect category	
	Major	Minor
III. STITCHING AND SEAM DEFECTS- (cont'd)		
Stitch tension:		
- tight stitch tension (stitches break when normal strain is applied in the direction of the seam or stitching)	116	
Open joining seam:		
- up to 1/4-inch (outside)		223
- 1/4-inch up to and including 1/2-inch (outside and inside)		224
- greater than 1/2-inch (outside and inside)	117	
Runoff, skipped or broken stitch:		
- joining seam – use open seam classification		225
- edge, top or raised stitching:		
- up to 1/4-inch	--	
- more than 1/2-inch	118	
IV. CUTTING		
Any component part not cut in accordance with specified pattern, directional lines on patterns or not in accordance with specification requirements	119	
Twill line of any outside part not running in specified direction (when applicable)	120	
V. COMPONENTS AND ASSEMBLY		
Any component part or operation omitted (unless otherwise specified herein)	121	
Any component not as specified (unless otherwise specified herein)	122	
Any component part tight, twisted, or distorted, pleated, or full (unless otherwise specified herein)	123	
Any operation not as specified (unless otherwise specified herein)	124	
Eyelet:		
- one or more omitted or added	125	
- chipped, broken, cracked, split, or bent	126	
- loose or raw edges exposed around edges of eyelet on outside	127	
- not securely clinched or clinched too tightly, cutting the fabric	128	
- positioned other than as specified	129	
- side or front puckered or pleated in clinching of one or more eyelets	130	
- raw edge of front stay lining exposed around edge of washer on inside	131	
Sweatband omitted:		
- short or tight, causing puckering, gathering or pleats on curtain corded edge of braid stitched through.		226
- end of sweatband not overlapped as specified, or not finished as specified		227

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TABLE I. End-item defects. – Continued

Defect Description	Defect category	
	Major	Minor
V. COMPONENTS AND ASSEMBLY - (cont'd)		
Sweatband:		
- seamed end of sweatband not spread open (where applicable)	133	
- finished width not as specified	134	
- seamed end of sweatband not turned under and stitched to crown side	135	
- joining seam of sweatband not positioned as specified from either side of back seam of service caps	136	
- upper edge of cloth sweatband not turned under and stitching to crown side	137	
- sweatband not of uniform width along major portion of sweatband	138	
- lower edge of sweatband exposed beyond rolled edge of curtain		228
- zig-zag stitching at overlapping end of sweatband catching any other part of the caps (when applicable)	139	
- lower edge of sweatband not positioned above bottom rolled edge of curtain as specified	140	
- folded end of band not on right or left side toward back of service caps	141	
- more than one splice		229
- raw edges of splice on outside of sweatband		230
- splice not sewn diagonally		231
Crown piece (top crown):		
- depth of fold at center not as specified	142	
- top edge poorly shaped or puckered	143	
- tapering at each end not as specified	144	
- width not as specified		233
- top crown seam offset from center:		
- 1/8-inch up to 1/4-inch inclusive		234
- more than 1/4-inch	145	
- joining to the side crown twisted, poorly shaped, tight, puckered or poorly constructed		235
- wider end of tip not at front of caps	146	
- crease line at center of tip not aligned with center front:		
- 1/8-inch to 1/4-inch inclusive		236
- more than 1/4-inch	147	
Crown pieces (side crown):		
- twisted, full, puckered or distorted	148	
- top edge poorly shaped or puckered	149	
- raise or top stitching on top seam omitted	150	
- side piece joining to curtain tight	151	
- point not fully forced out (unless otherwise specified herein)		237

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TABLE I. End item defects. – Continued

Defect Description	Defect category	
	Major	Minor
V. COMPONENTS AND ASSEMBLY - (cont'd)		
Crown pieces (side crown):		
- side-tip joining seam raised toward tip	152	
- tip exposed more than 1/8-inch above side	153	
- front and back seams of sides not spread open and top stitched		238
Curtain:		
- top edge poorly shaped	154	
- top edge long or stretched, causing curtain to stand away from side of service caps	155	
- top edge of curtain not positioned tightly against the cord edge of braid	156	
- curtain not covering back seam of crown as specified were required	157	
- overlap at front of service caps out of alignment with front seam by more than 1/4-inch	158	
- excessive fullness, puckering or distortion on side piece or on bottom edge of curtain		239
- overlap on front of service caps reversed, (i.e., right side overlapping left side, as worn)	159	
- point at center front where curtain overlaps, out of alignment with front seam by:		
- 1/4-inch to 1/2-inch, inclusive		240
- more than 1/2-inch	160	
- seam not properly forced out		241
- seam joining plies of curtain along top edge exposed on outside of service caps 1/16-inch or more for a distance of more than 1-inch where required	161	
- reverse lap at front (i.e., not overlapped as required)		242
- point of overlap of curtain at center front less than 3/8 or more than 5/8-inch below tip seam	162	
- front bottom edge not having rounded effect, i.e., pointed		243
- finished left end of curtain from center front of service caps not as specified	163	
- under ply of curtain exposed on front of curtain		244

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TABLE I. End item defects. – Continued

Defect Description	Defect category	
	Major	Minor
V. COMPONENTS AND ASSEMBLY - (cont'd)		
Braid on curtain:		
- tight, causing excessive puckering or fullness on curtain	164	
- corded edge irregular in width, or flat section of braid exposed beyond edge of curtain	165	
- loose, causing curtain not to lie smoothly	166	
Braid on curtain:		
- end of braid on short end of curtain extending into and caught in the seaming of curtain to bottom edge of crown side	167	
- edge frayed	168	
Lining and interlining:		
- tight, short, twisted, or distorted, causing any distortion on outside of service caps	169	
- overedge stitching along bottom edge of side piece and lining omitted	170	
- front and back seams of interlining not free from the side crown seams		245
- depth of tuck through center of tip less than 1/4 or more than 3/8-inch at center		246
- tuck through center of tip not tapered at each end	171	
- lining not anchored to tip-to-side joining seam	172	
- lining exposed below bottom edge of curtain		247
- any evidence of blee through, bubbling or color change	173	
- not properly adhered to basic material	174	
- omitted		248
Combination identification and size label:		
- missing, incorrect, or illegible	175	
- not securely caught in stitching		249
- not positioned as specified		250
- label stitching visible on side crown when curtain is turned up	176	
- size of lettering on label not as specified		251
- size marking not visible beyond edge of sweatband in finished service caps		252

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TABLE I. End item defects. – Continued

Defect Description	Defect category	
	Major	Minor
VI. PRESSING AND CLEANING EFFECTS		
Blocking or pressing omitted	177	
Poorly pressed (burned or scorched, impression or gloss on outside of caps, bottom edge of caps and curtain at front of back edge not pressed as specified)		253
Any spot or stain on outside visible at 3 feet (largest dimension in any direction):		
- up to 1/4-inch, inclusive	178	
- more than 1/4-inch		254
Any marking exposed or visible on outside		255
Thread ends:		
On inside:		
- three or more thread ends not trimmed to less than 1/4-inch		256
On outside:		
- More than one (1) but not more than three (3) not trimmed to less than 3/8-inch	179	
- More than three not trimmed to less than 3/8-inch		257
- three or more loose threads not removed		258
One or more shade or size tickets not removed		259
Re-sewn and/or re-stitched seams not cleaned		260
Thread end caught for a distance of more than 1/4-inch in a line of stitching on outside		261
Visible basting thread not removed		262

1/ Slubs and knots shall be rated in accordance with the Fabric Defect Replica Scales.
2/ Note minor defect 216 was deleted

3.7 Puckering. Refers to the gathering of a seam/stitches during sewing and after sewing causing unacceptable seam appearance when compared to the 68373A; AATCC Photographic Seam Smoothness Scales (see 2.3). Examine seams suspected of being puckered as specified in 4.2. Score as a puckered seam those major portions of the suspected seam that have a rating of less than “4” for single or double needle seams.

3.8 Quality assurance terms and procedures. Definitions of quality assurance terms and procedures, where applicable, are in accordance with ASQ/ANSI Z1.4.

3.9 Raw edge. Raw edge is defined as any edge not finished as specified including any cut edge of fabric when finishing of the edge is prescribed. A raw edge is prone to fraying and presents an untidy appearance on the outside of the service caps.

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3.10 Stitches and seams. Unless otherwise specified, define seams and stitching in accordance with ASTM D6193. Open seam defects are defined in 3.10.1.

3.10.1 Open seam. A seam is considered open when one or more stitches joining a seam are broken or runoff, or when two or more continuous skipped stitches occur. Double stitched seams are considered open when one or both rows of stitching are open.

4. GENERAL REQUIREMENTS

4.1 General. This section covers general aspects of style, format, and requirements for preparing a defense standard. This standard contains provisions and terminology for end-item inspection, examination, and dimensional examination for the service caps. If there are any inconsistencies between the requirements of the appropriate end-item specification and the provisions of this standard, the end-item specification shall govern.

4.2 Coverage. A standard shall be prepared to describe essential technical requirements for engineering or technical criteria, methods, processes, or practices. To the greatest extent possible, standards shall be written so that commercial practices and processes may be used to meet the requirement, and requirements should be stated in terms of desired outcomes as opposed to detailed procedures.

4.3 Normal viewing conditions. Normal viewing conditions for the inspection of the service caps is 3-feet from the surface being examined. Unless otherwise specified, the service caps shall be examined while lying flat on a suitable table using the following illumination criteria. The overhead direct lighting source shall be parallel to the viewing surface so as to illuminate with direct perpendicular impinging light rays. The surface illumination level shall be a minimum of 1075 lux (100-foot candles). The lights shall be cool white preheat rapid start fluorescent lamps with white reflectors and without baffles or glass.

5. DETAILED REQUIREMENTS

5.1 Sampling and acceptance criteria. The sample unit shall be one (1) finished service caps and lot size shall be expressed in the number of sample units. The sample size and Acceptance Quality Limits (AQLs) shall be as specified in the contract or purchase order. Sampling for inspection shall be conducted in accordance with ASQ/ANSI Z1.4 – Sampling Procedures and Tables for Inspection by Attributes. Sampling shall represent the production lot, and shall include samples from the beginning, middle and end of production of the end item lot.

5.2 End item dimensional examination. The finished service caps shall be examined for conformance to the finished measurements as stated in the end item specification. The sample size for end item dimensional examination shall be as specified in the contract or purchase order.

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5.3 End item visual examination. The finished service caps shall be examined for the defects stated in Table I of this document. Defects are assigned as “Major” or “Minor”. The sample unit shall be one (1) finished service caps.

5.4 Shade variation examination. During the examination, each service caps in the sample shall be examined for shade variation. Any service caps in the sample containing uneven shade or shade variation, within a part or between parts, shall be cause for rejection of entire lot represented by the sample.

5.4.1 Shade matching and appearance examination. The shade match and appearance examination shall be conducted as specified in the material specification, contract, or purchase order.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. To provide a standardized examination procedure for evaluating quality of service caps that are worn by members of the military. These defects are typical of those included in quality inspections of service caps. Non-Government standards should be used to describe the requirements for commercial interfaces, processes, practices, or methods.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.
b. Type of material defects allowed.
c. The specific issue of individual documents referenced (see 2.3).
d. Acceptance Quality Limits (AQLs).
e. Sample size.

6.3 Subject term (key word) listing.

Defect
Examination
Hat
Inspection
Uniforms

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

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Army – GL
Navy – NU
Air Force – 11

Preparing activity:
DLA - CT

(Project: 8405-2025-017)

Review activities:
Army – MD
Navy – MC

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